

CS & IT ENGINEERING



Computer Network

Flow Control

Lecture No. - 04



By - Abhishek Sir



Recap of Previous Lecture



Topic

Stop-and-Wait Protocol





Topics to be Covered



Topic

Stop-and-Wait Protocol



ABOUT ME



Hello, I'm **Abhishek**

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Topic : Cycle Time



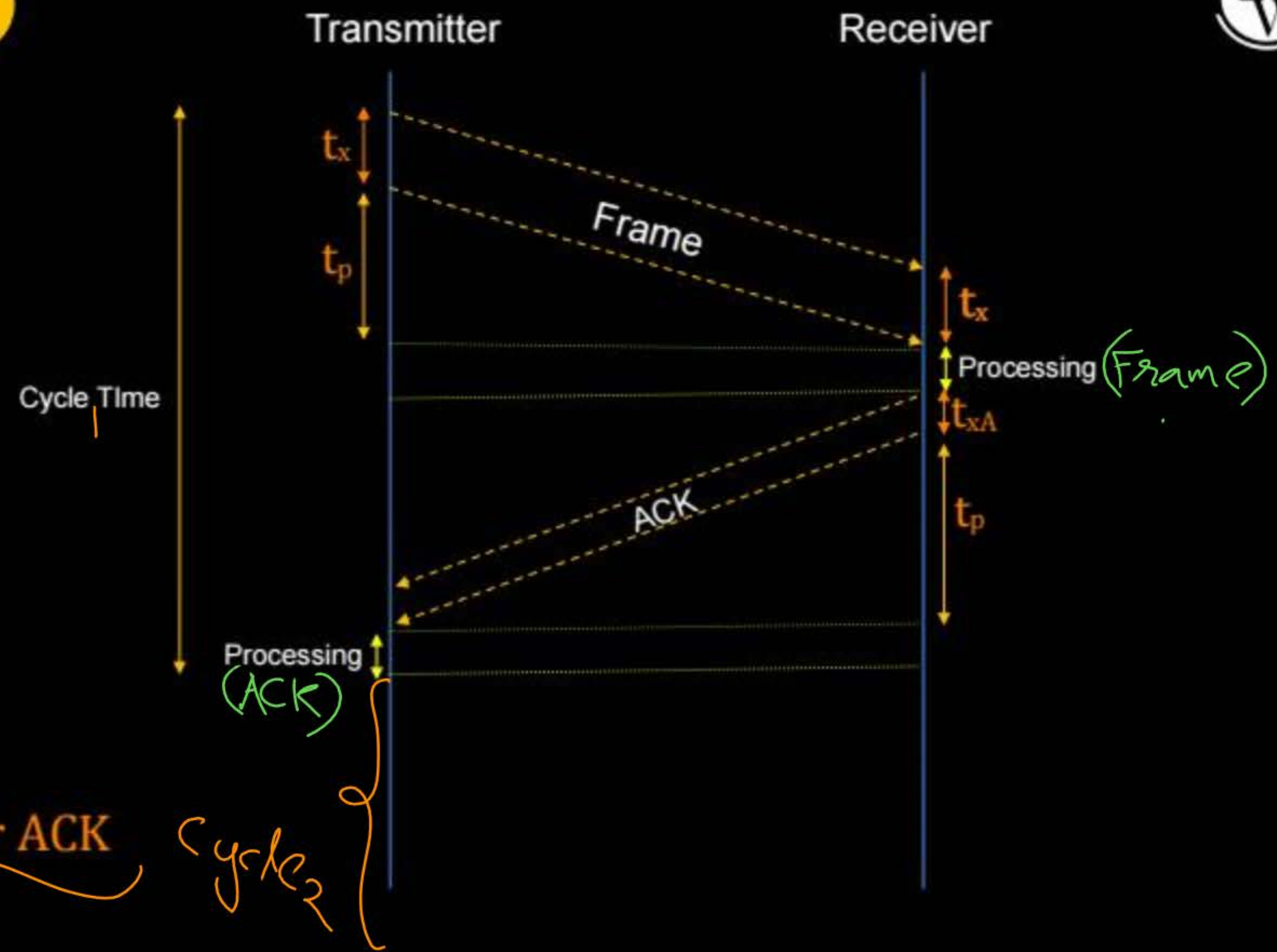
$$\begin{aligned} \text{Cycle time} = & \text{Transmission delay} + \text{Propagation delay} \\ & + \text{Queuing delay at receiver} + \text{Processing delay by receiver for frame} \\ & + \text{Transmission delay for ACK} + \text{Propagation delay} \\ & + \text{Queuing delay at transmitter} + \text{Processing delay by transmitter for ACK} \end{aligned}$$



Topic : Cycle Time



Time-out Value
 $\geq$ cycle time



t_{xA} : Transmission delay for ACK

cycle₂

$$t_{xA} \ll t_x$$



Topic : Cycle Time



$$\begin{aligned}\text{Cycle time} = & \text{Transmission delay} + \text{Propagation delay} \\ & + \text{Queuing delay at receiver} + \text{Processing delay by receiver} \\ & + \text{Transmission delay for ACK} + \text{Propagation delay} \\ & + \text{Queuing delay at transmitter} + \text{Processing delay by transmitter}\end{aligned}$$

Suppose queuing and processing delays are negligible at both the end.

$$\begin{aligned}\text{Cycle time} = & \text{Transmission delay} + \text{Propagation delay} \\ & + \text{Transmission delay for ACK} + \text{Propagation delay}\end{aligned}$$

$$= (t_x + t_p) + (t_{xA} + t_p)$$



Topic : Cycle Time



$$\text{Cycle time} = \text{Transmission delay} + \text{Propagation delay} \\ + \text{Transmission delay for ACK} + \text{Propagation delay}$$

Suppose Transmission delay for ACK is also negligible.

$$\text{Cycle time} = \text{Transmission delay} + \text{Round Trip Propagation delay}$$

$$= (t_x + 2t_p)$$



Topic : Round Trip Delay



For Data Link Layer :

$$\text{Round Trip Delay} / \text{Time (RTT)} = \text{Cycle time}$$

For Transport Layer :

$$\text{Round Trip Delay} / \text{Time (RTT)} = \text{Round Trip Propagation delay}$$

$\approx (2 * t_p)$



Topic : Efficiency



→ For Stop-and-Wait ARQ:

$$\text{Efficiency } (\eta) = \frac{\text{Transmission delay}}{\text{Cycle Time}}$$

$$\eta = 0 \text{ to } 1$$

$$\text{Efficiency } (\eta) = \frac{\text{Transmission delay}}{\text{Cycle Time}} * 100 \%$$

$$\eta = \frac{t_{\text{tx}}}{\text{cycle time}}$$

Example 5 :-

Consider two hosts A and B directly connected through point to point link using stop and wait ARQ for flow control. Suppose packet size is 1000 bytes, link bandwidth is 1 Mbps, distance is 2 Km and signal speed is 4 milisecond per kilometer. Calculate efficiency in percent (round off to nearest integer) ?



Solution:-

$$\text{Packet Size} = 1000 \text{ bytes} = 8 * 10^3 \text{ bits}$$

$$\text{Bandwidth} = 1 \text{ Mbps} = 10^6 \text{ bits / sec}$$

$$t_x = \frac{\text{Packet Size}}{\text{Bandwidth}} = \frac{8 * 10^3 \text{ bits}}{10^6 \text{ bits / sec}} = 8 * 10^{-3} \text{ sec} = 8 \text{ ms}$$

$$\text{Distance} = 2 \text{ Km}$$

$$\text{Signal Speed} = 4 \text{ ms/Km}$$

$$t_p = \text{Distance} * \text{Signal Speed} = 2 \text{ Km} * 4 \text{ ms/Km} = 8 \text{ ms}$$

$$\text{Cycle time} = (t_x + 2 * t_p) = 24 \text{ ms} = (8 + 2 * 8) \text{ ms}$$

For Stop-and-Wait ARQ:

$$\text{Efficiency } (\eta) = \frac{\text{Transmission delay}}{\text{Cycle Time}} * 100 \%$$

$$= \frac{8 \text{ ms}}{24 \text{ ms}} * 100 \%$$

$$= 33.33 \%$$

$$\approx 33 \%$$

$$\text{Ans} = 33 \text{ to } 34$$

#Q. Suppose we are transmitting frames between two nodes using Stop-and-Wait protocol. The frame size is 3000 bits. The transmission rate of the channel is 2000 bps (bits/second) and the propagation delay between the two nodes is 100 milliseconds. Assume that the processing times at the source and destination are negligible. Also, assume that the size of the acknowledgement packet is negligible. Which ONE of the following most accurately gives the channel utilization for the above scenario in percentage?

[GATE 2025, Set-2, 2-Mark]

IIT-R, H.W.

- (A) 88.23
- (B) 93.75
- (C) 85.44
- (D) 66.67

#Q. The values of parameters for the Stop-and-Wait ARQ protocol are as given below.

Bit rate of the transmission channel = 1Mbps

Propagation delay from sender to receiver = 0.75 ms

Time to process a frame = 0.25ms

Number of bytes in the information frame = 1980

Number of bytes in the acknowledge frame = 20

Number of overhead bytes in the information frame = 20

Assume that there are no transmission errors. Then the transmission efficiency (expressed in percentage) of the Stop-and-Wait ARQ protocol for the above parameters is _____ (correct to 2 decimal place).

[GATE 2017, Set-1, 2-Mark]

IIT-R, H.W

#Q. Suppose two hosts are connected by a point-to-point link and they are configured to use Stop-and-Wait protocol for reliable data transfer. Identify in which one of the following scenarios, the utilization of the link is the lowest.

[GATE 2023, 1-Mark]

1/T - R, H.W.

- A** Longer link length and lower transmission rate
- B** Longer link length and higher transmission rate
- C** Shorter link length and lower transmission rate
- D** Shorter link length and higher transmission rate



Topic : Efficiency



→ To achieve 100% utilization ($\eta = 1$) in Stop-and-Wait ARQ

$$\text{Efficiency } (\eta) = \frac{\text{Transmission delay}}{\text{Cycle Time}}$$

$$\text{Cycle Time} = \text{Transmission delay}$$

$$(t_x + 2t_p) = t_x$$

→ 100% utilization ($\eta = 1$) in Stop-and-Wait ARQ
[Only when propagation delay and other latency are Zero]



Topic : Efficiency



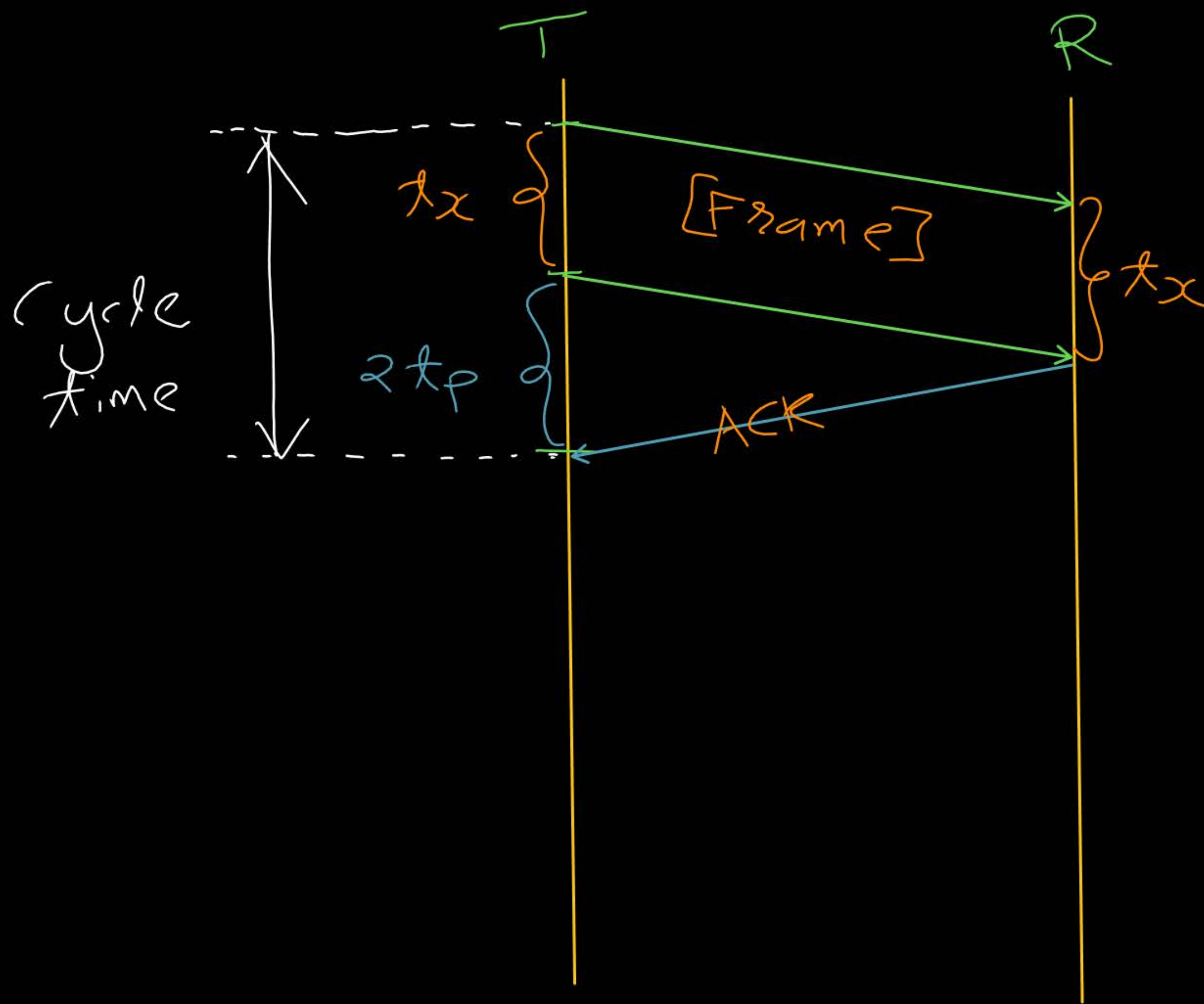
→ To achieve 50% utilization ($\eta = 1/2$) in Stop-and-Wait ARQ

$$\text{Efficiency } (\eta) = \frac{\text{Transmission delay}}{\text{Cycle Time}}$$

$$\text{Cycle Time} = 2 * \text{Transmission delay}$$

$$(t_x + 2t_p) = 2 * t_x$$

$$t_x = 2 t_p$$



Example 6 :-

Consider two hosts A and B directly connected through point to point link using stop and wait ARQ for flow control. Suppose link bandwidth is 2 Mbps and one-way propagation delay is 8 milisecond. To achieve a link utilization of at least 50% the minimum frame size is _____ bytes.



Solution:-

$$\text{Bandwidth} = 2 \text{ Mbps} = 2 * 10^6 \text{ bits / sec}$$

$$t_p = 8 \text{ ms} = 8 * 10^{-3} \text{ Sec}$$

To achieve 50% utilization in Stop-and-Wait ARQ.

$$\text{Cycle time} = 2 * t_x$$

$$(t_x + 2 * t_p) = 2 * t_x$$

$$t_x = 2 * t_p$$

$$\text{Frame Size} = (2 * t_p) * \text{Bandwidth}$$

$$= (2 * 8 * 10^{-3} \text{ Sec}) * (2 * 10^6 \text{ bits / sec})$$

$$= 32 * 10^3 \text{ bits}$$

$$= 4 * 10^3 \text{ bytes}$$

$$\text{Ans} = 4000$$

#Q. A channel has a bit rate of 4 kbps and one-way propagation delay of 20 ms. The channel uses stop and wait protocol. The transmission time of the acknowledgement frame is negligible. To get a channel efficiency of at least 50%, the minimum frame size should be

[GATE 2005]

IIT-B, H.W.

- A** 80 bytes
- B** 80 bits
- C** 160 bytes
- D** 160 bits

#Q. Suppose that the stop-and-wait protocol is used on a link with a bit rate of 64 kilobits per second and 20 milliseconds propagation delay. Assume that the transmission time for the acknowledgment and the processing time at nodes are negligible. Then the minimum frame size in bytes to achieve a link utilization of at least 50% is _____.

[GATE 2015, Set-1, 2-Mark]

IIT-K, H.W



Topic : Efficiency



→ To achieve 25% utilization ($\eta = 1/4$) in Stop-and-Wait ARQ

$$\text{Efficiency } (\eta) = \frac{\text{Transmission delay}}{\text{Cycle Time}}$$

$$\text{Cycle Time} = 4 * \text{Transmission delay}$$

$$(t_x + 2t_p) = 4 t_x$$

$$t_x = \frac{2}{3} t_p$$

#Q. A link has a transmission speed of 10^6 bits/sec. It uses data packets of size 1000 bytes each. Assume that the acknowledgment has negligible transmission delay, and that its propagation delay is the same as the data propagation delay. Also assume that the processing delays at nodes are negligible. The efficiency of the stop-and-wait protocol in this setup is exactly 25%. The value of the one-way propagation delay (in milliseconds) is _____.

[GATE 2015, Set-2, 2-Mark]

||T-K, H.W.



Topic : Channel Utilization

→ Link Utilization or Throughput

$$\text{Throughput} = \frac{[\text{Packet Size}]}{[\text{Cycle Time}]}$$

$$= \frac{[\text{Packet Size}]}{[\text{Bandwidth}]} \times \text{Bandwidth}$$

cycle time

① Efficiency

① a) 0 to 1

① b) percent (0 to 100)

② Utilization

→ Throughput

(Bytes per sec)

$$\text{Throughput} = \text{Efficiency} * \text{Data Transfer Rate at sender}$$

Example 7 :-

Consider two hosts A and B directly connected through point to point link using stop and wait ARQ for flow control. Suppose packet size is 50 bytes, link bandwidth is 2 Kbps and one-way propagation delay is 100 milisecond, the sender throughput is _____ bytes/second?

Ans = 125

Solution:-

$$\text{Packet Size} = 50 \text{ bytes} = 400 \text{ bits}$$

$$\text{Bandwidth} = 2 \text{ Kbps} = 2 * 10^3 \text{ bits / sec}$$

$$t_x = \frac{\text{Packet Size}}{\text{Bandwidth}} = \frac{400 \text{ bits}}{2 * 10^3 \text{ bits / sec}} = 200 \text{ ms}$$

$$t_p = 100 \text{ ms}$$

$$\text{Cycle time} = (t_x + t_p) + (t_{xA} + t_p) = 400 \text{ ms}$$

$$= (200 + 100) + (0 + 100) \text{ ms}$$

$$\text{Throughput} = \frac{\text{Packet Size}}{\text{Cycle Time}} = \frac{[50 \text{ bytes}]}{[400 \text{ ms}]} = \frac{50 \times 10^3}{400} \text{ bytes/sec}$$

$$= \underline{125 \text{ bytes / sec}}$$

$$\eta = \frac{t_x}{\text{cycle time}} = \frac{200 \text{ ms}}{400 \text{ ms}} = \frac{1}{2}$$

$$\begin{aligned} \text{Throughput} &= \eta * \text{Bandwidth} = \frac{1}{2} * 2 \text{ Kbps} = 1 \text{ Kbps} \\ &= 10^3 \text{ bits/sec} = \frac{10^3}{8} \text{ bytes/sec} = 125 \text{ Bytes/sec} \end{aligned}$$

#Q. A sender uses the Stop-and-Wait ARQ protocol for reliable transmission of frames. Frames are of size 1000 bytes and the transmission rate at the sender is 80 Kbps (1Kbps = 1000 bits/second). Size of an acknowledgement is 100 bytes and the transmission rate at the receiver is 8 Kbps. The one-way propagation delay is 100 milliseconds. Assuming no frame is lost, the sender throughput is _____ bytes/second.

[GATE 2016, Set-1, 2-Mark]

11SC, H.W.

#Q. On a wireless link, the probability of packet error is 0.2. A stop-and-wait protocol is used to transfer data across the link. The channel condition is assumed to be independent from transmission to transmission. What is the average number of transmission attempts required to transfer 100 packets?

[GATE 2006]

IIT-KGP
H.W

A 100

B 125

C 150

D 200



2 mins Summary



Topic

Stop-and-Wait ARQ ✓



THANK - YOU

